difficulty? Well, although the answer is indeed yes, this reasoning is too glib and misses the real intent of the question.

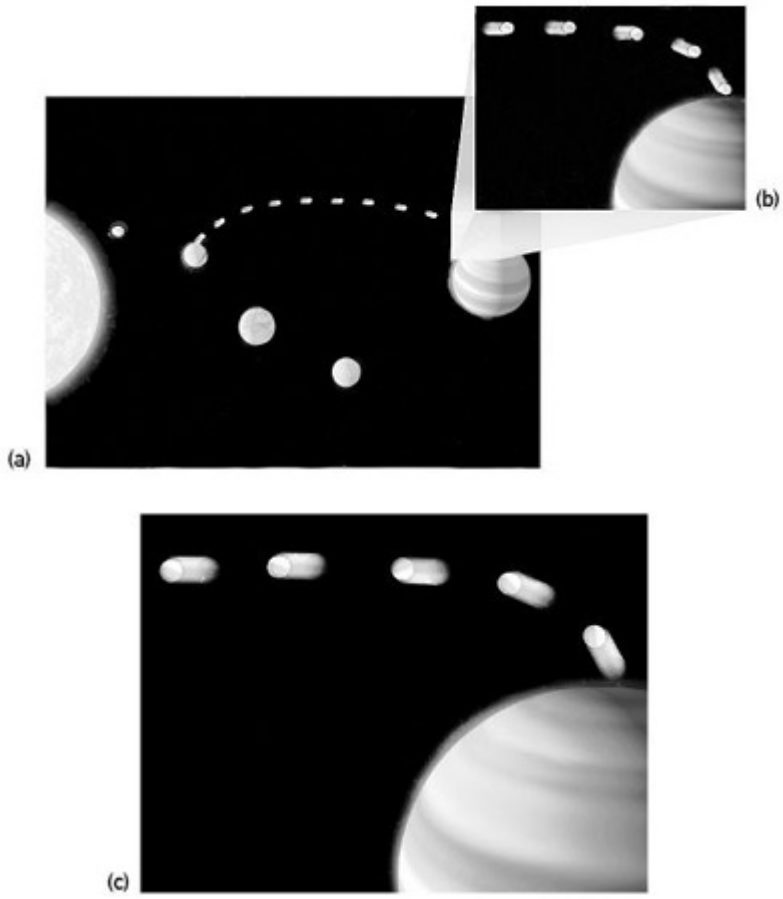


Figure 6.1 (a) A tennis ball flying from Venus to Jupiter together with (b) a close-up. (c) Tennis ball's motion if its velocity is reversed just before it hits Jupiter.

When you run the film in reverse, you see the tennis ball leap from Jupiter's surface, moving upward and toward the left, with exactly the same speed (but in exactly the opposite direction) from when it hit the planet. This initial part of the film is certainly consistent with the laws of physics: we can imagine, for example, someone launching the tennis ball from Jupiter's surface with precisely this velocity. The essential question is whether the *rest* of the reverse run is also consistent with the laws of physics. Would a ball launched with this initial velocity—and subject to Jupiter's downward-pulling gravity—actually move along the trajectory depicted in the rest of the reverse run film? Would it exactly retrace its original downward trajectory, but in reverse?